

“BODIES OF MICROSCOPICALLY VISIBLE SIZE SUSPENDED IN A LIQUID WILL PERFORM MOVEMENTS OF SUCH A MAGNITUDE THAT THEY CAN BE... OBSERVED IN A MICROSCOPE.”

Albert Einstein, on Brownian motion, July 18, 1905

ALBERT EINSTEIN PUBLISHED FOUR REVOLUTIONARY PAPERS

in what became known as his Annus Mirabilis (Miracle Year). His first expanded Max Planck's idea from 1900 that energy existed in minute packets of energy. Einstein proposed that **packets of light energy** could help explain the **photoelectric effect**, in which wavelength (not intensity) of light provides the energy needed to eject electrons from a metal surface—later proven experimentally (see 1921).

Einstein's second theory explained how the random movement of tiny particles in gas or water was caused by the motion of molecules bombarding the particles—Brownian motion. In September, **Einstein published his theory of special relativity** (see pp.244–45). In it

671 MILLION THE SPEED OF LIGHT IN MILES PER HOUR

he reconciled the constancy of the speed of light with the principle of relativity: the idea that mechanical processes happened in the same way whether at rest or moving. Light had previously been regarded as an exception to this principle—but Einstein used an analysis of

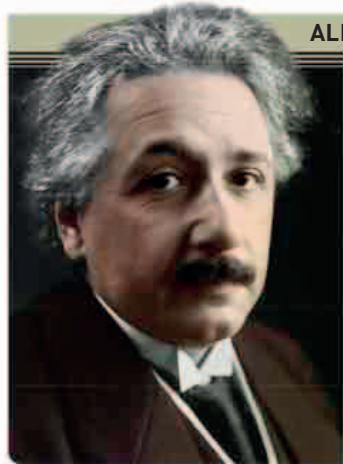
time and space to show that it was compatible. Finally in November, he **published a conclusion** that arose from his work on relativity. He suggested that when an object emitted energy, it lost mass too, so energy and mass were interchangeable. This relationship was expressed in a **simple equation: $E = mc^2$** .

British biologist **William Bateson** (1861–1926) was among a number of scientists interested in the study of inheritance. In a letter of April this year he called it **genetics**.

Growing populations needed more food, so the demand for crop fertilizer increased. German chemists devised a way to make ammonia, the compound that provided nitrogen for plant growth. Ammonia is formed from hydrogen and nitrogen.

Fritz Haber (1868–1934) described a key reaction between atmospheric nitrogen and hydrogen. During World War I, when natural sources of nitrate came under Allied control, **Carl Bosch** (1874–1940) used Haber's principle to produce industrial quantities of ammonia.

German chemist **Alfred Einhorn** (1856–1917) succeeded in making the local anesthetic **procaine**, later traded as **Novocaine**. The drug would become a standard painkiller.



ALBERT EINSTEIN (1879–1955)

Born in Germany, Einstein took Swiss citizenship before finding work at the Bern patent office in 1903. Here he wrote his ground-breaking papers and was awarded a doctorate by the University of Zurich. He completed his general theory of relativity in 1915 and was awarded the Nobel Prize in 1922. In later years, Einstein became a US citizen.



Albert Calmette (center) studied animal toxins and developed some of the first antivenoms. Later he collaborated with Camille Guérin to make BCG vaccine.

BRITISH GEOLOGIST RICHARD OLDHAM (1858–1936) studied the seismic shocks that went through Earth following earthquakes. At the end of the previous century Oldham had identified two kinds of waves: fast-moving, longitudinal, primary P-waves and slower, transverse, secondary S-waves. He found that below a certain depth within Earth, **seismic**

waves travel much more slowly—and S-waves stop altogether. In February, Oldham suggested this phenomenon could be explained by the **presence of a core inside Earth** that was composed of a different kind of material. Later studies would show that Earth has a core located at a depth of 1,802 miles (2,900 km), and confirmed that the outer core is fluid.

On Christmas Eve, a Canadian inventor, working at the US Weather Bureau, **broadcast the first radio program**. Reginald Fessenden (1866–1932), a rival of Guglielmo Marconi, transmitted his program from Brant Rock, Massachusetts. It included a voice message and music was directed to ships in the Atlantic Ocean more used to receiving messages in Morse code.

At the Institut Pasteur, France, work began on a program that led to the development of a **new vaccine** that would protect millions of peoples from a **deadly disease: tuberculosis (TB)**. French scientists **Albert Calmette** (1863–1933) and **Camille Guérin** (1872–1961) had been inspired by the historical uses of the harmless cowpox as a way to vaccinate humans against dangerous smallpox (see 1796). They thought that a similar process could be tried



TUBERCULOSIS (TB)

German physician Robert Koch (1843–1910) identified the bacteria that caused TB, and was awarded the Nobel Prize in 1905. Formerly called consumption, TB was often fatal. It infects the lungs—causing lesions, or tubercles—and is spread through droplets dispersed by coughs and sneezes. The BCG vaccine was the first protection against TB.

May William Bateson coins the term genetics

Fritz Haber publishes The Thermodynamics of describing a reaction to make ammonia

July 18 Einstein publishes his theory of Brownian motion

September 26 Einstein publishes his theory of special relativity

November 21 Einstein explains the relationship between mass and energy: $E = mc^2$

June 9 Albert Einstein publishes his theory of the photoelectric effect

October American zoologist Edmund Wilson describes the XX and XY system of sex chromosomes

Danish astronomer Ejnar Hertzsprung distinguishes between giant and dwarf stars

January Arthur Eddington begins a statistical study of drifting stars

February 13 US Patent Office registers a local anesthetic procaine for Alfred Einhorn

German astronomer Max Wolf discovers the first Trojan asteroid, named 588 Achilles

February 21 Thomas Oldham suggests Earth has a core, later shown to be liquid

German physicist Karl Schwarzschild explains the observed darkening of a star from center to edge